Step 11 Continuous Abstractness Report

Score Diagnostics

Metric	Value
Questions with score	30
Raw score mean	0.6333
Raw score SD	0.3801
Raw score median	0.6666
Raw score min	0.0000
Raw score max	1.0000
Binary cutoff used	0.50
Questions below cutoff	15
Questions above cutoff	15
Cutoff reproduces binary label	True

Variable
Total word cour
Sentence count
Average sentenc
ENEM item cor
Mean response

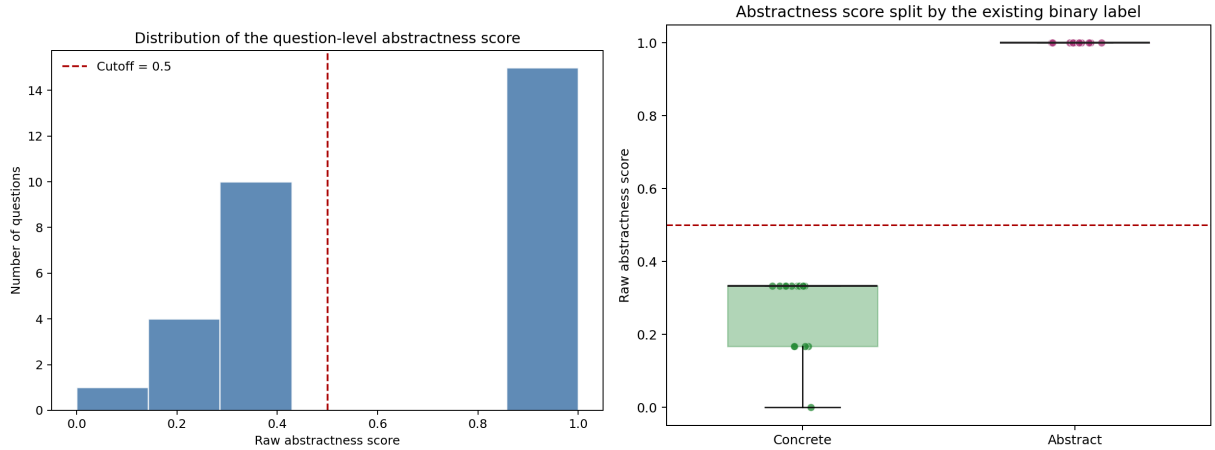

Figure 1: Distribution of the raw question-level abstractness score and its split by the existing binary label.

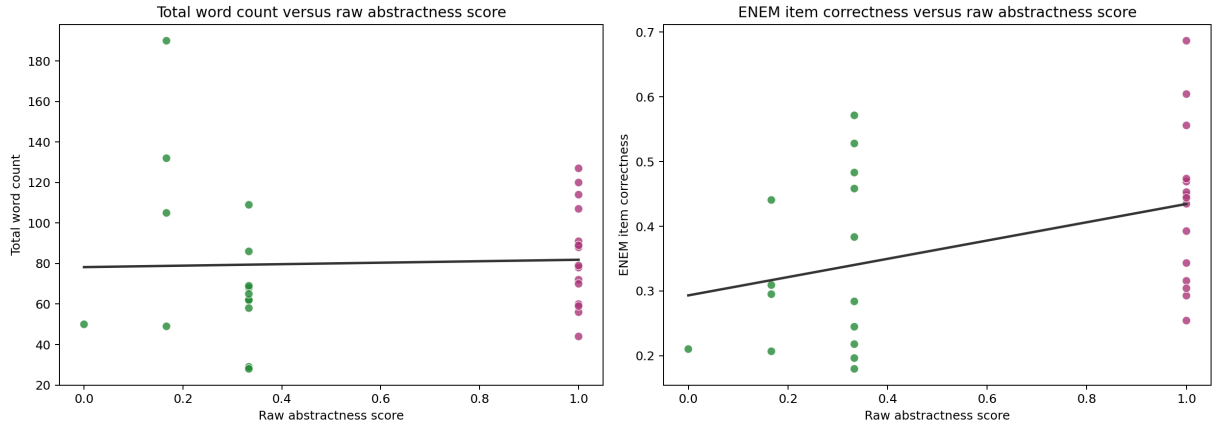

Figure 2: Question-level abstractness score against two key item properties retained in the regression models.

Primary Continuous-Score Model

Metric	Value
Robust SE type	Two-way cluster (participant + question)
Included participants	36
Included questions	30
Included trials	1075
Abstractness beta	1.311e-08
Abstractness SE	1.251e-08
Abstractness statistic	1.048
Abstractness p-value	0.303503
Abstractness 95% CI	[-1.249e-08, 3.870e-08]
Global block-order beta	4.586e-09
Within-block-position beta	-6.080e-09
Word-count beta	-4.518e-08
ENEM-correctness beta	3.554e-09

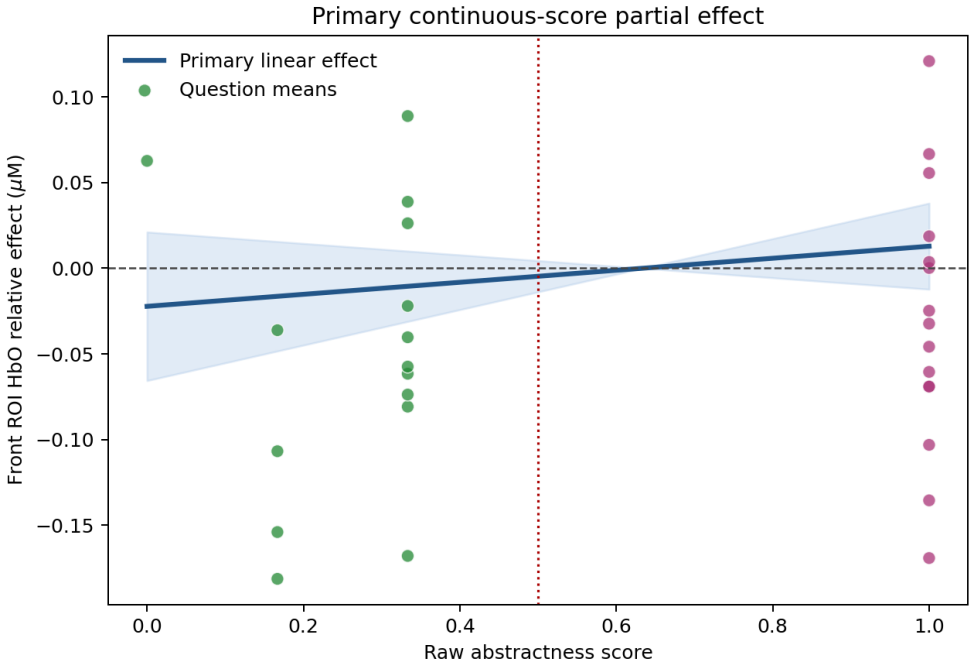

Figure 3: Primary continuous-score partial-effect visualization. Higher values indicate more abstract questions.

Benchmarks And Sensitivity Checks

Model	Target term	Beta	SE	p	95 Binary abstract-versus-concrete benchmark model
Binary label	3.016e-08	2.351e-08	0.209749	[-1.793e-08, 7.824e-08]	
Quadratic continuous abstractness model	z(Abstractness score)	1.628e-08	1.208e-08	0.188168	[-8.426e-09, 4.099e-08]
Quadratic continuous abstractness model	Squared score term	2.722e-08	2.164e-08	0.218468	[-1.704e-08, 7.149e-08]
Threshold-strength benchmark model	Distance from cutoff	1.559e-08	2.307e-08	0.504510	[-3.159e-08, 6.276e-08]
Back ROI continuous abstractness benchmark	z(Abstractness score)	5.166e-09	6.101e-09	0.404042	[-7.311e-09, 1.764e-08]
Fixed-window continuous abstractness benchmark	z(Abstractness score)	1.162e-08	1.036e-08	0.271618	[-9.583e-09, 3.281e-08]
No-override continuous abstractness model	z(Abstractness score)	8.210e-09	1.152e-08	0.481741	[-1.535e-08, 3.177e-08]

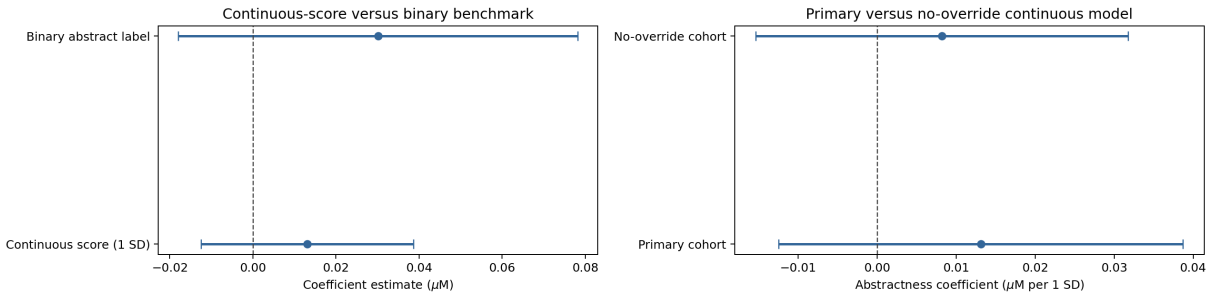

Figure 4: Comparison against the binary benchmark and the no-override sensitivity model.

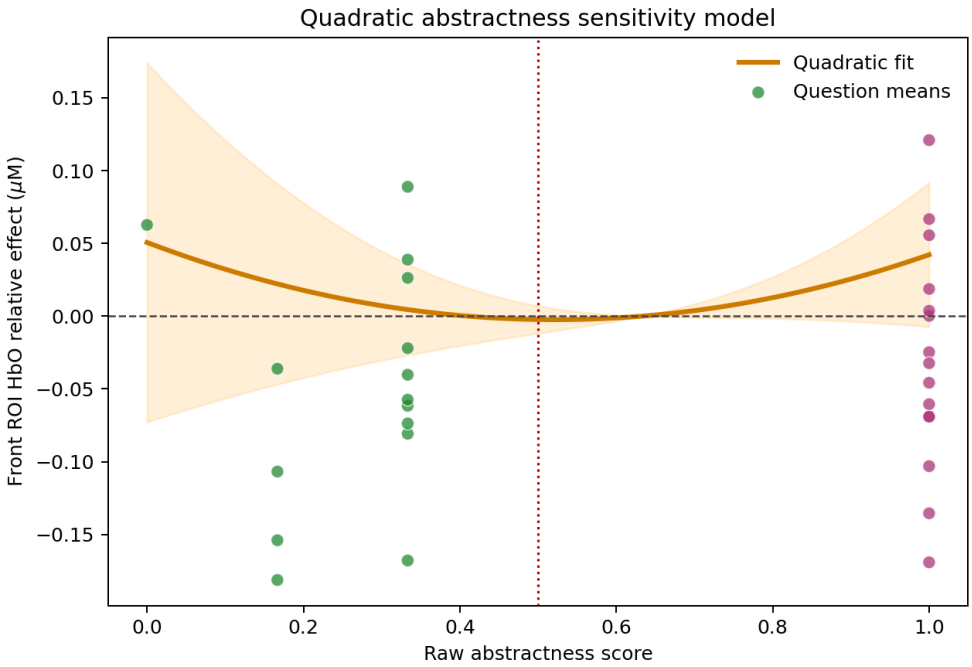

Figure 5: Quadratic sensitivity model across the raw abstractness score.

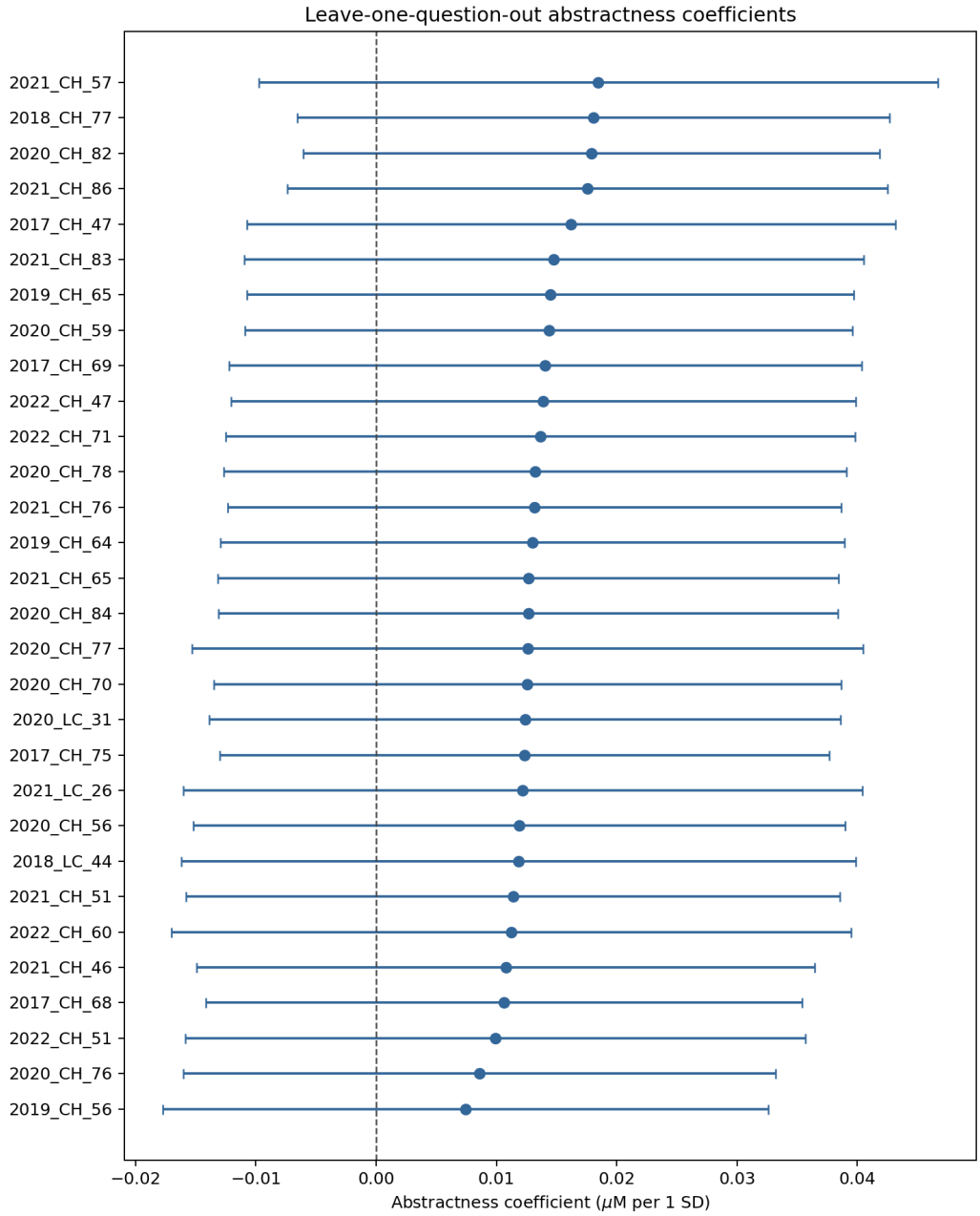

Figure 6: Leave-one-question-out abstractness coefficients with 95% confidence intervals.

Final Conclusion

In the participant-fixed-effects regression using the continuous abstractness score, the association between abstractness level and left frontal ROI HbO response was not statistically significant (beta=1.311e-08, SE=1.251e-08, p=0.303503, 95